The V_n Invariants as Colored Links–Gould Invariants: A Root–Center Approach

Jiuhe Liu

Abstract

The knot invariants V_n arise from a rank-two Nichols algebra and a $4n$-dimensional right Yetter–Drinfeld module, whereas the colored Links–Gould invariants are defined from typical $U_q(\mathfrak{sl}(2|1))$-modules. We prove that these two constructions agree, up to the mirror and parameter inversion forced by the right-module convention. The proof is structural. We first realize the braid operator T_n as a gauge transform of a canonical super Yetter–Drinfeld braiding. We then construct the relevant right–right paired double, prove that its Hopf pairing is perfect in every root degree, and obtain its completed universal R-matrix. An explicit Abelian Drinfeld twist separates this double into an all-odd $U_\hbar(\mathfrak{sl}(2|1))$ root factor and a commutative central factor. Under this factorization, the Nichols module becomes a typical all-odd highest-weight module tensored with a one-dimensional central module. Finally, we transport duality, all four oriented crossings, partial transposes, and writhe normalization. For every oriented knot $\mathcal{K}$, every $n \geq 1$, and every $\beta \neq 0, -1$, the result is

$$V_{n,\mathcal{K}}\big(Q^{2n\beta+n}, Q^2\big) = LG^{(n)}_{\mathcal{K}}\big(Q^{-n\beta}, Q^{-1}\big) = LG^{(n)}_{\overline{\mathcal{K}}}\big(Q^{n\beta}, Q\big)\,.$$

In particular, the scalar-identity property conjectured for the endomorphism-valued V_n construction follows from the simplicity of the corresponding typical module.

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1 Introduction

The invariants V_n of Garoufalidis and Kashaev are extracted from a family of braid-form Yang–Baxter operators T_n acting on $4n$-dimensional spaces Y_n. The construction is rooted in a rank-two Nichols algebra and is strikingly explicit. The colored Links–Gould invariant, by contrast, is built from typical modules over a quantum deformation of $\mathfrak{sl}(2|1)$. The two theories therefore begin with rather different algebraic input. The purpose of this paper is to explain why they nevertheless produce the same knot invariant.

The connection is not obtained by a direct comparison of two large matrices. Instead, we expose a sequence of structural identifications:

$$\begin{aligned}&\text{Nichols braid operator } T_n \longleftrightarrow \text{super Yetter–Drinfeld braiding}\\ &\longleftrightarrow \text{braiding of a completed paired double}\\ &\xrightarrow{\text{root–center twist}} U_\hbar^{\Pi_{\mathrm{ao}}}(\mathfrak{sl}(2|1))\widehat{\otimes}\mathcal{Z}_{\mathrm{cen}}.\end{aligned}$$

The $4n$-dimensional object splits at the last stage as

$$\mathcal{Y}_n \cong W_{n,\beta}^{\mathrm{ao}} \boxtimes L_{n,\beta},$$

where $W_{n,\beta}^{\mathrm{ao}}$ is a simple all-odd highest-weight module and $L_{n,\beta}$ is one-dimensional and central. The central line contributes a computable crossing scalar. It disappears on balanced normal long-knot diagrams and, more generally, under the standard writhe normalization.

There are two points at which a braid-level comparison could fail to produce a knot invariant. First, the Hopf pairing used to form the double must be well defined, nondegenerate in the relevant completion, and strong enough to produce a genuine universal R-matrix. We prove this using explicit alternating-word bases and root-degree pairing matrices. Second, a conjugacy on $Y_n \otimes Y_n$ does not by itself control cups, caps, mixed orientations, partial transposes, or curls. We therefore transport the entire rigid tangle calculus and compare the transported duality maps with the ordinary finite-dimensional ones.

1.1 Related work and scope

The original Links–Gould invariant was constructed from the four-dimensional family of typical modules over $U_q(\mathfrak{gl}(2|1))$, using a tensor-state model for $(1,1)$-tangles [1]. This is an instance of the Reshetikhin–Turaev construction from a ribbon category [2]. Because typical modules for Lie superalgebras have vanishing ordinary quantum dimension, the natural closed-link theory is more conveniently expressed using a renormalized invariant or modified trace. For $\mathfrak{sl}(2|1)$, this point of view was developed by Geer and Patureau–Mirand [12], and it underlies the modern colored Links–Gould family.

The relevant algebraic background has several distinct layers. The classification and root theory of basic classical Lie superalgebras go back to Kac [3], while changes of Borel through isotropic odd roots belong to the generalized-root-system framework of Serganova [4]. Quantized enveloping superalgebras and their universal R-matrices were constructed explicitly by Yamane [5]. Etingof–Kazhdan quantization in the super setting and its relation to Drinfeld–Jimbo presentations, including arbitrary Cartan matrices, were studied by Geer [10, 9]. Geer's comparison with the Kontsevich integral identifies the corresponding quantum-group and weight-system invariants for Lie superalgebras of type A–G [11]. These general results supply essential structural background, but they do not by themselves identify the particular right–right completed double, module gauges, duality conventions, and writhe normalization used below; those interfaces are treated explicitly in this paper.

On the Nichols-algebra side, the spaces Y_n and their braid operators were introduced by Garoufalidis and Kashaev in their construction of multivariable knot polynomials from braided Hopf algebras with automorphisms [6]. Garoufalidis, Harper, Kashaev, Kohli, and Wagner subsequently defined the colored Links–Gould sequence from $4n$-dimensional typical modules, established its cabling and modified trace framework, and formulated the equality with the V_n-sequence as a conjecture; in that work the comparison was proved for $n = 2$ [8, Conjecture 1.4 and Theorem 1.5]. Thus the contribution claimed here is not the definition of either family. It is the proposed all-n structural bridge: the realization of the Nichols operator in a completed paired double, its root–center separation, the identification of the coloring object, and the transport of the full oriented rigid tangle data needed to compare the normalized knot scalars.

We work with a formal quantum parameter, away from the atypical values $\beta = 0, -1$. The comparison concerns normalized oriented-knot invariants, computed from normal $(1,1)$-tangles; it does not assert an equivalence of the entire module categories. The completion conventions are specified in Sections 2.1 and 3.2.

1.2 Main results

Let $\Bbbk$ be a field of characteristic zero,

$$R = \Bbbk[[\hbar]], \qquad \mathbb{K} = \Bbbk((\hbar)), \qquad Q = e^{\hbar}.$$

The invariant $LG_{\mathcal{K}}^{(n)}(q^{\alpha}, q)$ is normalized using the $4n$-dimensional typical distinguished-Borel module $V_0(n-1, \alpha)$, with value 1 on the unknot.

Theorem 1.1 (Main theorem)**.** *For every $n \geq 1$, every $\beta \in \Bbbk \setminus \{0, -1\}$, and every oriented knot $\mathcal{K}$,*

$$V_{n,\mathcal{K}}\left(Q^{2n\beta+n}, Q^2\right) = LG_{\mathcal{K}}^{(n)}\left(Q^{-n\beta}, Q^{-1}\right) = LG_{\overline{\mathcal{K}}}^{(n)}\left(Q^{n\beta}, Q\right). \tag{1}$$

Moreover, the endomorphism-valued long-knot invariant defining V_n is a scalar multiple of the identity on $\mathcal{Y}_n$.

The algebraic mechanism behind Theorem 1.1 is summarized by the next two statements.

The proof has two algebraic ingredients. The completed paired double has a root-degree perfect Hopf pairing and a universal R-matrix (Theorem 3.3); an Abelian twist separates its root and central Hopf factors (Theorem 4.1 and Proposition 4.2). Under this separation, the coloring object has the module decomposition above (Theorem 5.1). The braid comparison extends to oriented crossings and duality, yielding the long-knot transport formula of Theorem 7.1.

1.3 Outline of the proof

Section 2 constructs the positive Hopf superalgebra and its finite-dimensional super Yetter–Drinfeld module. Section 3 constructs the completed paired double and its universal R-matrix. Section 4 separates the root and central factors, and Section 5 identifies the coloring module, including its distinguished-Borel highest weight. Section 6 compares all oriented crossings and duality maps. Section 7 assembles these local identities into long-knot transport and proves the comparison with the standard colored Links–Gould invariant. Appendix A checks the original Garoufalidis–Kashaev conventions and records the parameter dictionary; Appendix B gives the first root-degree terms of the canonical tensor.

2 The Nichols module and its braiding

2.1 Conventions

Unless otherwise stated, algebras, modules, pairings, and linear maps are over $\mathbb{K}$. Formal quantum groups and exponentials are first defined over R, with tensor products of their R-forms $\hbar$-adically completed, and are then extended to $\mathbb{K}$. In particular, $\mathbb{K}\widehat{\otimes}_R U_\hbar$ means $\mathbb{K} \otimes_R U_\hbar$ with the R-form $U_\hbar$ already complete; no further ideal-adic completion after inverting $\hbar$ is intended. Completed tensor products of the resulting $\mathbb{K}$-algebras are likewise formed from the indicated completed R-forms before extension of scalars. The field $\mathbb{K}$ carries its Laurent series valuation topology, not the ideal-adic topology of $(\hbar) \subset \mathbb{K}$, since $\hbar$ is invertible. Infinite root-degree products use the additional directed completion specified in Section 3.2. For an even element H,

$$Q^H = e^{\hbar H}.$$

If a, b, c, d are homogeneous elements of super vector spaces, then

$$(a \otimes b)(c \otimes d) = (-1)^{|b||c|} ac \otimes bd. \tag{2}$$

Every module in this paper is a right module. Thus, if $\rho(X)(v) = v \triangleleft X$, then

$$\rho(XY) = \rho(Y)\rho(X). \tag{3}$$

Our universal R-matrix convention is

$$\mathcal{R}\Delta(a) = \Delta^{\mathrm{op}}(a)\mathcal{R}. \tag{4}$$

It follows from (3) that the canonical braiding on right modules is

$$c^{\mathcal{R}}_{V,W} = \tau^{\mathrm{s}}\rho_{V\otimes W}(\mathcal{R}^{-1}), \tag{5}$$

where τ^{s} is the super flip. This inverse is responsible for the mirror in Eq. (1).

2.2 The reduced positive half

Let $\mathcal{H}^+$ be the topological Hopf superalgebra over $\mathbb{K}$ obtained by extending scalars from the $\hbar$-adic completion of the R-superalgebra generated by even elements H_1, H_2 and odd elements x_1, x_2, subject to

$$[H_1, H_2] = 0, \qquad [H_i, x_j] = a_{ij}x_j, \qquad x_1^2 = x_2^2 = 0, \qquad A = (a_{ij}) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{6}$$

No additional quadratic relation is imposed between x_1x_2 and x_2x_1. The Hopf structure is

$$\begin{aligned} \Delta(H_i) &= H_i \otimes 1 + 1 \otimes H_i, \quad \varepsilon(H_i) = 0, \quad S(H_i) = -H_i, \\ \Delta(x_i) &= x_i \otimes 1 + Q^{H_i} \otimes x_i, \quad \varepsilon(x_i) = 0, \quad S(x_i) = -Q^{-H_i}x_i. \end{aligned} \tag{7}$$

2.3 The space $\mathcal{Y}_n$

Let $B_{\mathbb{K}}$ be the superalgebra generated by x_1, x_2 with only $x_1^2 = x_2^2 = 0$. A bihomogeneous element of bidegree (a, b) has parity $\overline{a+b}$. For nonzero bihomogeneous elements $x, y \in B_{\mathbb{K}}$ with $\deg x = (a, b)$ and $\deg y = (c, d)$, define

$$\chi(x, y) = (-1)^{(a+b)(c+d)}Q^{ad+bc}. \tag{8}$$

The full diagonal braiding is the linear map determined by

$$\tau_\chi(x \otimes y) = \chi(x, y)y \otimes x.$$

In particular,

$$\chi(x, x_1) = (-1)^{a+b}Q^b, \qquad \chi(x, x_2) = (-1)^{a+b}Q^a.$$

We use the same notation on the bihomogeneous elements of $\mathcal{Y}_n \subset B_{\mathbb{K}}$ defined below. All subsequent uses of χ and all Sweedler formulas are understood on bihomogeneous components. Fix $n \geq 1$ and set

$$t = Q^{2n\beta+n}. \tag{9}$$

Define

$$v_n = (x_1x_2)^n + t(x_2x_1)^n.$$

For $\beta \neq 0, -1$, let $\mathcal{Y}_n$ be the span of

$$\begin{aligned} &\{1, v_n\} \cup \{(x_1x_2)^k, (x_2x_1)^k : 1 \leq k < n\} \\ &\qquad \cup \{(x_1x_2)^k x_1, (x_2x_1)^k x_2 : 0 \leq k < n\}. \end{aligned} \tag{10}$$

This displayed set is a basis, so $\dim_{\mathbb{K}} \mathcal{Y}_n = 4n$.

The parameter and convention dictionary identifying $\mathcal{Y}_n$ with the right Yetter–Drinfeld module Y_n of [6, Proposition 7.4] is given in Appendix A. Under this identification, T_n denotes precisely the braid-form R-matrix defined by [6, Theorem 3.6 and Definition 7.5], after extension of scalars to $\mathbb{K}$.

2.4 The super Yetter–Drinfeld structure

Introduce the group-like element

$$U_n = Q^{-2n\beta H_1 + 2n(\beta+1)H_2}. \tag{11}$$

For homogeneous $y \in \mathcal{Y}_n$ of bidegree (a, b), define

$$\begin{aligned} y \triangleleft H_1 &= -by, & y \triangleleft H_2 &= -ay, \\ y \triangleleft x_1 &= yx_1 - Q^{-2n(\beta+1)}\chi(y, x_1)x_1 y, \\ y \triangleleft x_2 &= yx_2 - Q^{2n\beta}\chi(y, x_2)x_2 y. \end{aligned} \tag{12}$$

The boundary identities

$$v_n \triangleleft x_1 = v_n \triangleleft x_2 = 0, \qquad ((x_1x_2)^{n-1}x_1) \triangleleft x_2 = v_n, \qquad ((x_2x_1)^{n-1}x_2) \triangleleft x_1 = t^{-1}v_n$$

show directly that the action preserves $\mathcal{Y}_n$.

For y of bidegree (a, b), write the braided coproduct in $B_{\mathbb{K}}$ as

$$\Delta_B(y) = \sum y^{(1)} \otimes y^{(2)}, \qquad \text{with bihomogeneous summands.}$$

Then

$$\delta_n^{\mathrm{s}}(y) = \sum (-1)^{|y^{(1)}||y^{(2)}|}\chi(y^{(2)}, y^{(1)})^{-1} y^{(1)} \otimes U_n Q^{-aH_1 - bH_2} y^{(2)}. \tag{13}$$

The explicit parity sign in (13) cancels the Koszul sign in $\chi(y^{(2)}, y^{(1)})^{-1}$; the super tensor category already supplies the Koszul interchange.

Proposition 2.1. *Equations* (12) *and* (13) *make* $\mathcal{Y}_n$ *a right–right super Yetter–Drinfeld module over* $\mathcal{H}^+$.

Proof. For y of bidegree (a, b), set

$$\iota_n(y) = U_n Q^{-aH_1 - bH_2} y \in \mathcal{H}^+.$$

The action (12) is the pullback along ι_n of the right super-adjoint action

$$A \triangleleft_{\mathrm{ad}} h = \sum (-1)^{|A||h_{(1)}|} S(h_{(1)}) A h_{(2)}.$$

Moreover,

$$\Delta(\iota_n(y)) = (\iota_n \otimes \mathrm{id})\delta_n^{\mathrm{s}}(y).$$

Thus $\iota_n(\mathcal{Y}_n)$ is stable under the right adjoint action and is a right coideal. The standard adjoint/coideal calculation, with the Koszul rule (2), gives the right–right super Yetter–Drinfeld identity. Counitality and coassociativity follow by restricting those of Δ. □

The identification of the original operator T_n with the braiding transported from this module is established in Section 6; the parameter and action conventions are checked in Appendix A.

3 The completed paired double

3.1 The negative half and the Hopf pairing

Define $\mathcal{H}^-$ using even generators $\widehat{H}_1, \widehat{H}_2$, and odd generators F_1, F_2, with

$$[\widehat{H}_1, \widehat{H}_2] = 0, \qquad [\widehat{H}_i, F_j] = a_{ji} F_j, \qquad F_i^2 = 0,$$

and

$$\begin{aligned} \Delta(\widehat{H}_i) &= \widehat{H}_i \otimes 1 + 1 \otimes \widehat{H}_i, \qquad S(\widehat{H}_i) = -\widehat{H}_i, \\ \Delta(F_i) &= F_i \otimes Q^{\widehat{H}_i} + 1 \otimes F_i, \quad S(F_i) = -F_i Q^{-\widehat{H}_i}. \end{aligned} \tag{14}$$

The even Hopf pairing $\eta : \mathcal{H}^- \widehat{\otimes} \mathcal{H}^+ \to \mathbb{K}$ is determined by

$$\eta(\widehat{H}_i, H_j) = -\frac{a_{ji}}{\hbar}, \qquad \eta(F_i, x_j) = \delta_{ij}, \tag{15}$$

all mixed Cartan–root pairings being zero, and by the right-dual order

$$\eta(fg, h) = \sum \eta(g, h_{(1)}) \eta(f, h_{(2)}), \qquad \eta(f, hk) = \sum \eta(f_{(1)}, h) \eta(f_{(2)}, k), \tag{16}$$

with the Koszul sign understood in the second formula.

3.2 Alternating-word bases, the Hopf pairing, and completion

For $m \geq 1$, let $X_{i,m}$ be the alternating word of length m in x_1, x_2 beginning with x_i, and let $F_{i,m}$ be its negative analogue. The adjacent-square relations generate monomial ideals, so the words that avoid adjacent squares form bases of the root algebras. These are exactly the alternating words. Their degree pieces are

$$B_m^+ = \mathbb{K} X_{1,m} \oplus \mathbb{K} X_{2,m}, \qquad B_m^- = \mathbb{K} F_{1,m} \oplus \mathbb{K} F_{2,m}.$$

Set $B_0^+ = B_0^- = \mathbb{K}1$, and let $\mathcal{C}^+$ and $\mathcal{C}^-$ be the Cartan subalgebras generated by H_1, H_2 and $\widehat{H}_1, \widehat{H}_2$, respectively: their polynomial R-forms are $\hbar$-adically completed before extension to $\mathbb{K}$, as specified in Section 2.1. The commuting Cartan generators act diagonally on the alternating words. The corresponding commuting weight derivations preserve the adjacent-square ideals, so each half is the semidirect product of its root algebra with its Cartan algebra. Thus placing the Cartan coefficients on the left gives unique normal forms and topological vector-space decompositions

$$\mathcal{H}^\pm \cong \widehat{\bigoplus}_{m \geq 0} \mathcal{C}^\pm B_m^\pm.$$

Lemma 3.1. *The formulas (15)–(16) descend uniquely to a continuous even Hopf pairing on $\mathcal{H}^- \widehat{\otimes} \mathcal{H}^+$, and $\eta(B_m^-, B_l^+) = 0$ for $m \neq l$.*

Proof. In the free Hopf superalgebras, the odd–odd terms in the coproduct cancel by the Koszul rule, so the square relations on either side annihilate the relation ideal on the other. Primitive Cartan elements preserve Cartan commutativity, while $\eta(Q^{\widehat{H}_i}, Q^{H_j}) = Q^{-a_{ji}}$ preserves the fixed-weight Cartan–root relations. Hence both relation ideals lie in the radicals of the free pairing, and the pairing descends. Root-degree orthogonality follows from the skew-primitive coproducts. □

The Hopf pairing defines the right–right paired double

$$\mathfrak{D}_{\mathrm{rr},\eta} = \mathcal{H}^- \bowtie_\eta \mathcal{H}^+.$$

The cross relations in the double are

$$\begin{aligned} [H_i, \widehat{H}_j] &= 0, & [H_i, F_j] &= -a_{ij} F_j, \\ [\widehat{H}_i, x_j] &= -a_{ji} x_j, & x_i F_j + F_j x_i &= \delta_{ij}(Q^{H_i} - Q^{\widehat{H}_i}). \end{aligned} \tag{17}$$

Put $P_r(Q) = \prod_{s=1}^{r}(1 - Q^{-2s})$, with $P_0(Q) = 1$, and let

$$M_m(i,j) = \eta(F_{i,m}, X_{j,m}).$$

Proposition 3.2 (Root-degree perfectness)**.** *For $r \geq 0$ and $r \geq 1$, respectively,*

$$\begin{aligned} M_{2r+1} &= (-1)^r P_r(Q) I_2, \\ M_{2r} &= (-1)^r P_{r-1}(Q) \begin{pmatrix} 1 & -Q^{-r} \\ -Q^{-r} & 1 \end{pmatrix}. \end{aligned} \tag{18}$$

Consequently,

$$\det M_{2r+1} = P_r(Q)^2, \qquad \det M_{2r} = P_{r-1}(Q) P_r(Q),$$

and $\eta|_{B_m^- \otimes B_m^+}$ is perfect for every m.

Proof. Apply (16) recursively to the last letter of each alternating word. The two possible starting letters give a two-state recurrence. At odd length its off-diagonal entries cancel, while at even length the two possible terminal pairings differ by Q^{-r}. Starting from $M_1 = I_2$ and

$$M_2 = \begin{pmatrix} -1 & Q^{-1} \\ Q^{-1} & -1 \end{pmatrix}$$

gives (18) by induction. Taking determinants and using $P_r = P_{r-1}(1 - Q^{-2r})$ proves the claim. Every $P_r(Q)$ is nonzero in $\mathbb{K} = \Bbbk((\hbar))$. □

We use the directed root-degree completion

$$\widehat{\mathcal{T}}_{+,-} = \prod_{m \geq 0} (\mathcal{C}^+ B_m^+) \widehat{\otimes}_{\mathbb{K}} (\mathcal{C}^- B_m^-). \tag{19}$$

The product completes only the common root degree; in each factor, the Cartan tensor product is formed from the completed R-forms before extension to $\mathbb{K}$. Root degrees add under multiplication, so every fixed degree of a product contains only finitely many summands.

Theorem 3.3 (Completed universal R-matrix)**.** *The signed right-dual coevaluation series of the paired double $\mathfrak{D}_{\mathrm{rr},\eta}$ belongs to (19) and is invertible there. Define $\mathcal{R}_{\mathrm{rr}}$ to be its inverse. Then*

$$\mathcal{R}_{\mathrm{rr}} \Delta(d) = \Delta^{\mathrm{op}}(d) \mathcal{R}_{\mathrm{rr}}, \tag{20}$$

$$(\Delta \otimes \mathrm{id})(\mathcal{R}_{\mathrm{rr}}) = \mathcal{R}_{13}\mathcal{R}_{23}, \qquad (\mathrm{id} \otimes \Delta)(\mathcal{R}_{\mathrm{rr}}) = \mathcal{R}_{13}\mathcal{R}_{12}. \tag{21}$$

Proof. Proposition 3.2 supplies a dual basis in each root degree. The super right-dual coevaluation contributes the additional sign $(-1)^m$ in degree m. The resulting series belongs to (19), and its inverse is defined recursively degree by degree. Dualizing multiplication and coproduct through (16) gives

$$\begin{aligned}\Delta(d)\mathcal{R}_{\mathrm{rr}}^{-1} &= \mathcal{R}_{\mathrm{rr}}^{-1}\Delta^{\mathrm{op}}(d),\\ (\Delta\otimes\mathrm{id})(\mathcal{R}_{\mathrm{rr}}^{-1}) &= \mathcal{R}_{23}^{-1}\mathcal{R}_{13}^{-1},\\ (\mathrm{id}\otimes\Delta)(\mathcal{R}_{\mathrm{rr}}^{-1}) &= \mathcal{R}_{12}^{-1}\mathcal{R}_{13}^{-1}.\end{aligned}$$

Inverting these identities proves (20) and (21). No identification of either half with the full continuous dual of the other is required. □

The double action induced from the Yetter–Drinfeld structure agrees, through (5), with the canonical braiding c_n^{s}.

4 Root–center factorization

4.1 Root and central variables

Define

$$\mathsf{h}_i = \frac{H_i - \widehat{H}_i}{2}, \qquad \mathsf{c}_i = H_i + \widehat{H}_i. \tag{22}$$

The c_i are central. Normalize the odd root vectors by

$$E_i = Q^{-\mathsf{c}_i/4}x_i, \qquad G_i = Q^{-\mathsf{c}_i/4}F_i. \tag{23}$$

This separates the multiplication but not yet the coproduct.

4.2 The Abelian twist

Because $A^{-1} = A$, define

$$\mathcal{F}_{\mathrm{rc}} = \exp\left[\frac{\hbar}{4}\sum_{i,j=1}^{2}(A^{-1})_{ji}(\mathsf{h}_i\otimes\mathsf{c}_j - \mathsf{c}_j\otimes\mathsf{h}_i)\right]. \tag{24}$$

All entries in the exponent are commuting even primitive elements. Therefore $\mathcal{F}_{\mathrm{rc}}$ is a normalized Drinfeld twist. The twisted paired double has coproduct

$$\Delta_{\mathrm{rc}}(a) = \mathcal{F}_{\mathrm{rc}}\Delta(a)\mathcal{F}_{\mathrm{rc}}^{-1}.$$

Its universal R-matrix is obtained by twisting the universal R-matrix $\mathcal{R}_{\mathrm{rr}}$ of Theorem 3.3:

$$\mathcal{R}_{\mathrm{rc}} = \mathcal{F}_{\mathrm{rc},21}\mathcal{R}_{\mathrm{rr}}\mathcal{F}_{\mathrm{rc}}^{-1}. \tag{25}$$

Thus $\mathcal{R}_{\mathrm{rc}}$ is the universal R-matrix associated with Δ_{rc}, on the same underlying algebra.

For the convention $\Delta_F = F\Delta F^{-1}$, the twisted antipode is $S_F = u_F S u_F^{-1}$, where $u_F = m(\mathrm{id}\otimes S)(F)$. In the present case the Cartan elements in (24) are commuting, even, and primitive, so

$$m(\mathrm{id}\otimes S)(\mathcal{F}_{\mathrm{rc}}) = \exp\left[\frac{\hbar}{4}\sum_{i,j=1}^{2}(A^{-1})_{ji}(-\mathsf{h}_i\mathsf{c}_j + \mathsf{c}_j\mathsf{h}_i)\right] = 1.$$

Consequently,

$$S_{\mathrm{rc}} = S. \tag{26}$$

Thus the twist leaves the dual-module actions unchanged, and the canonical duality comparison for the twist equivalence is the identity. The tensorator itself is generally nontrivial, but its contribution contracts to the identity in evaluation and coevaluation. This twist therefore introduces no additional duality gauge.

Theorem 4.1 (Root–center separation)**.** *After twisting, the root and central Hopf structures separate:*

$$\mathfrak{D}_{\mathrm{rr},\eta}^{\mathcal{F}_{\mathrm{rc}}} \cong \mathcal{D}_{\mathrm{root}} \widehat{\otimes} \mathcal{Z}_{\mathrm{cen}}. \tag{27}$$

Moreover,

$$\mathcal{D}_{\mathrm{root}} \cong \mathbb{K} \widehat{\otimes}_R U_\hbar^{\Pi_{\mathrm{ao}}}(\mathfrak{sl}(2|1)).$$

Here the standard positive and negative Chevalley generators correspond to E_i and $(Q - Q^{-1})^{-1} G_i$, respectively, and the Cartan generators correspond to h_i.

For clarity, the complete root presentation used in this theorem is

$$\begin{gathered} [\mathsf{h}_i, \mathsf{h}_j] = 0, \qquad [\mathsf{h}_i, E_j] = a_{ij} E_j, \qquad [\mathsf{h}_i, G_j] = -a_{ij} G_j, \\ E_1^2 = E_2^2 = G_1^2 = G_2^2 = 0, \qquad E_i G_j + G_j E_i = \delta_{ij} (Q^{\mathsf{h}_i} - Q^{-\mathsf{h}_i}). \end{gathered} \tag{28}$$

No additional quadratic relation is imposed between $E_1 E_2$ and $E_2 E_1$, or between $G_1 G_2$ and $G_2 G_1$. Every central generator commutes with every generator in (28).

Proof. Conjugating the four skew-primitive terms by $\mathcal{F}_{\mathrm{rc}}$ gives

$$\begin{aligned} \Delta_{\mathrm{rc}}(E_i) &= E_i \otimes 1 + Q^{\mathsf{h}_i} \otimes E_i, \\ \Delta_{\mathrm{rc}}(G_i) &= G_i \otimes Q^{-\mathsf{h}_i} + 1 \otimes G_i. \end{aligned}$$

The $\mathsf{h}_i, \mathsf{c}_i$ remain primitive, so their exponentials remain group-like. Thus no central generator occurs in the coproduct of a root generator. Conversely, the c_i and their exponentials generate a central commutative Hopf algebra. The multiplication relations from (17) become the all-odd $\mathfrak{sl}(2|1)$ relations, with both simple roots isotropic and Cartan matrix A. To check that the algebra map is an isomorphism, use the inverse substitutions

$$H_i = \mathsf{h}_i + \tfrac{1}{2}\mathsf{c}_i, \qquad \widehat{H}_i = -\mathsf{h}_i + \tfrac{1}{2}\mathsf{c}_i, \qquad x_i = Q^{\mathsf{c}_i/4} E_i, \qquad F_i = Q^{\mathsf{c}_i/4} G_i.$$

Centrality of the c_i and (28) recover all Cartan–root commutators and square relations. They also give

$$x_i F_j + F_j x_i = \delta_{ij} Q^{\mathsf{c}_i/2} (Q^{\mathsf{h}_i} - Q^{-\mathsf{h}_i}) = \delta_{ij} (Q^{H_i} - Q^{\widehat{H}_i}),$$

recovering (17). Hence the inverse substitutions define an algebra map inverse to the generator change (22)–(23). The coproduct formulas above make this a Hopf isomorphism, proving (27). □

4.3 Factorization of the universal R-matrix

We now describe how $\mathcal{R}_{\mathrm{rc}}$ separates under the Hopf isomorphism (27). Let $\mathcal{R}^{\mathrm{rc}}_{\mathrm{root}}$ denote the inverse of the signed canonical root tensor after replacing alternating x- and F-words by the corresponding E- and G-words. This is the root-vector factor; the complete universal R-matrix of $\mathcal{D}_{\mathrm{root}}$ also includes its Cartan factor. Define the root and central Cartan factors by

$$\mathcal{R}^{\mathrm{root}}_{\mathrm{Cart}} = \exp[-\hbar(\mathsf{h}_1 \otimes \mathsf{h}_2 + \mathsf{h}_2 \otimes \mathsf{h}_1)]\,,$$
$$\mathcal{R}^{\mathrm{cen}}_{\mathrm{Cart}} = \exp\left[\frac{\hbar}{4}(\mathsf{c}_1 \otimes \mathsf{c}_2 + \mathsf{c}_2 \otimes \mathsf{c}_1)\right].$$

Proposition 4.2. *In the directed completion,*

$$\mathcal{R}_{\mathrm{rc}} = \mathcal{R}^{\mathrm{rc}}_{\mathrm{root}}\mathcal{R}^{\mathrm{root}}_{\mathrm{Cart}}\mathcal{R}^{\mathrm{cen}}_{\mathrm{Cart}}. \tag{29}$$

Under (27), the product $\mathcal{R}^{\mathrm{rc}}_{\mathrm{root}}\mathcal{R}^{\mathrm{root}}_{\mathrm{Cart}}$ belongs to the completed tensor square of $\mathcal{D}_{\mathrm{root}}$ and is its universal R-matrix, while $\mathcal{R}^{\mathrm{cen}}_{\mathrm{Cart}}$ belongs to the completed tensor square of $\mathcal{Z}_{\mathrm{cen}}$ and is its universal R-matrix. Their embeddings into the completed tensor square of the twisted double commute, and their product is $\mathcal{R}_{\mathrm{rc}}$. Thus (27) separates the quasitriangular structures as well as the Hopf algebra structures.

Proof. Substitution of $H_i = \mathsf{h}_i + \frac{1}{2}\mathsf{c}_i$ and $\widehat{H}_i = -\mathsf{h}_i + \frac{1}{2}\mathsf{c}_i$ into the untwisted Cartan exponential yields pure-root, mixed, and pure-central terms. Since the exponent of $\mathcal{F}_{\mathrm{rc}}$ is antisymmetric, $\mathcal{F}_{\mathrm{rc},21} = \mathcal{F}_{\mathrm{rc}}^{-1}$; the twist factors in (25) cancel the mixed terms exactly. They simultaneously turn every root tensor $X_{i,m} \otimes F_{j,m}$ into $E_{i,m} \otimes G_{j,m}$. This proves (29). The universal R-identities follow by twisting Theorem 3.3. Under (27), apply the Hopf projections $\mathrm{id} \otimes \varepsilon_{\mathrm{cen}}$ and $\varepsilon_{\mathrm{root}} \otimes \mathrm{id}$ to both tensor legs of these identities. The first projection removes the central Cartan factor, leaving $\mathcal{R}^{\mathrm{rc}}_{\mathrm{root}}\mathcal{R}^{\mathrm{root}}_{\mathrm{Cart}}$; the second removes the root factors, leaving $\mathcal{R}^{\mathrm{cen}}_{\mathrm{Cart}}$. Hence each is a universal R-matrix for the corresponding Hopf factor. Centrality gives the asserted commutation. □

Consequently, on modules that split into root and central factors, the braiding splits into the corresponding two contributions.

5 Identification of the coloring module

5.1 The root and central factors

On a homogeneous $y \in \mathcal{Y}_n$ of degree (a, b), the root and central Cartan elements act by

$$\begin{aligned} y \triangleleft \mathsf{h}_1 &= (n(\beta+1) - b)y, \quad & y \triangleleft \mathsf{h}_2 &= -(n\beta + a)y, \\ y \triangleleft \mathsf{c}_1 &= -2n(\beta+1)y, & y \triangleleft \mathsf{c}_2 &= 2n\beta y. \end{aligned} \tag{30}$$

Let $W^{\mathrm{ao}}_{n,\beta}$ be the simple all-odd right highest-weight module with even highest vector $w_{n,\beta}$ satisfying

$$w_{n,\beta} \triangleleft E_1 = w_{n,\beta} \triangleleft E_2 = 0, \qquad w_{n,\beta} \triangleleft \mathsf{h}_1 = n\beta w_{n,\beta}, \qquad w_{n,\beta} \triangleleft \mathsf{h}_2 = -n(\beta+1)w_{n,\beta}. \tag{31}$$

Let $L_{n,\beta}$ be the one-dimensional central module with characters

$$\mathsf{c}_1 \mapsto -2n(\beta+1), \quad \mathsf{c}_2 \mapsto 2n\beta.$$

5.2 Module identification and the braiding factors

Theorem 5.1 (Module identification)**.** *For $\beta \neq 0, -1$,*

$$\mathcal{Y}_n \cong W^{\mathrm{ao}}_{n,\beta} \boxtimes L_{n,\beta} \tag{32}$$

as a right module over the twisted paired double. The root factor $W^{\mathrm{ao}}_{n,\beta}$ is simple and has dimension $4n$.

Proof. By (12) and (30), the nonzero even vector v_n is killed by E_1, E_2 and has the highest weight (31). Let $M = v_n \triangleleft \mathcal{D}_{\mathrm{root}} \subseteq \mathcal{Y}_n$. As a cyclic highest-weight module, M has the simple highest-weight quotient $W^{\mathrm{ao}}_{n,\beta}$. The dimension formula (35), established independently in Section 5.3 below, and the basis (10) give

$$4n = \dim_{\mathbb{K}} W^{\mathrm{ao}}_{n,\beta} \leq \dim_{\mathbb{K}} M \leq \dim_{\mathbb{K}} \mathcal{Y}_n = 4n.$$

Thus $M = \mathcal{Y}_n$ and the quotient map is an isomorphism of root modules. Finally, the central characters in (30), together with Theorem 4.1, supply the factor $L_{n,\beta}$, proving (32). □

Fixing a nonzero vector in $L_{n,\beta}$, the identification (32) induces a root-module isomorphism

$$\Phi_n : \mathcal{Y}_n \xrightarrow{\sim} W^{\mathrm{ao}}_{n,\beta}.$$

The factorization (29) now gives two contributions to the braiding. Define the root braiding by

$$R'_{n,\beta} = c^{\mathrm{ao}}_{W^{\mathrm{ao}}_{n,\beta}, W^{\mathrm{ao}}_{n,\beta}}.$$

By (5), it is obtained from the inverse of the root universal R-matrix. The inverse of the central Cartan factor acts on $L_{n,\beta} \otimes L_{n,\beta}$, and hence on the central factors of $\mathcal{Y}_n \otimes \mathcal{Y}_n$, by the scalar

$$\lambda^{\mathrm{cen}}_{n,\beta} = Q^{2n^2\beta(\beta+1)}. \tag{33}$$

Indeed, substituting the characters $\mathsf{c}_1 = -2n(\beta+1)$ and $\mathsf{c}_2 = 2n\beta$ into $(\mathcal{R}^{\mathrm{cen}}_{\mathrm{Cart}})^{-1}$ gives this value.

5.3 Odd reflection and typicality

We now determine the dimension of the abstract simple module $W^{\mathrm{ao}}_{n,\beta}$, independently of $\mathcal{Y}_n$. Set $\alpha = n\beta$ and reflect the all-odd simple system at the first isotropic root. The distinguished Cartan data are

$$\gamma_1 = \alpha_1 + \alpha_2, \qquad \gamma_2 = -\alpha_1, \qquad h^{\mathrm{dist}}_1 = \mathsf{h}_1 + \mathsf{h}_2, \qquad h^{\mathrm{dist}}_2 = -\mathsf{h}_1.$$

Write $w = w_{n,\beta}$. The relation (28) gives

$$(w \triangleleft G_1) \triangleleft E_1 = (Q^{n\beta} - Q^{-n\beta})w.$$

Thus $w^{\mathrm{dist}} = w \triangleleft G_1 \neq 0$ when $\beta \neq 0$. The odd-reflection calculation [4, 5] makes this an odd highest vector for the reflected Borel. After the right-to-left conversion, its weights are

$$h^{\mathrm{dist}}_1 \cdot w^{\mathrm{dist}} = (n-1)w^{\mathrm{dist}}, \qquad h^{\mathrm{dist}}_2 \cdot w^{\mathrm{dist}} = n\beta w^{\mathrm{dist}}.$$

Here the conversion is $a \cdot u = (-1)^{|a||u|} u \triangleleft S(a)$ for homogeneous a, u; in particular it negates the Cartan weights. If $V_0(m, \alpha)$ denotes the simple distinguished highest-weight module with an even highest vector, highest-weight classification gives

$$W^{\mathrm{ao}}_{n,\beta} \longleftrightarrow \Pi V_0(n-1, n\beta). \tag{34}$$

For $m \in \mathbb{Z}_{\geq 0}$, the distinguished module $V_0(m, \alpha)$ is typical when $\alpha(\alpha + m + 1) \neq 0$, and then has dimension $4(m+1)$. These facts follow from the classical typicality criterion and character formula in [12, §1.1, equation (2)], together with the character-preserving formal quantization recalled in [12, §1.2]; see also [8, §3.3, equation (39), and Lemma 3.4] for the quantum-module formulation. We use these formulas after extension to $\mathbb{K}$, with $Q = e^{\hbar}$, so there is no root-of-unity specialization. For $m = n-1$ and $\alpha = n\beta$, the typicality condition is

$$\alpha(\alpha + m + 1) = n^2 \beta(\beta + 1) \neq 0.$$

Consequently $W^{\mathrm{ao}}_{n,\beta}$ is finite dimensional and

$$\dim_{\mathbb{K}} W^{\mathrm{ao}}_{n,\beta} = 4n \qquad (\beta \neq 0, -1). \tag{35}$$

The parity shift in (34) records our even all-odd highest vector; the associated tangle conventions are compared in Section 7.

6 Oriented crossings and duality

6.1 Integer-colored braid gauges and transported duality

We first identify the positive crossing T_n with a gauge transform of the canonical super Yetter–Drinfeld braiding. The same gauge extends to dual strands and cups and caps; Section 6.2 compares the transported maps with standard partial transposes.

Fix $\beta \in \Bbbk \setminus \{0, -1\}$. We construct a single gauge for braid generators and oriented rigid transport with arbitrary integer colors. Positive colors label the modules constructed above, negative colors label their left categorical duals, and zero labels the tensor unit.

For $n \geq 1$, write

$$\mathcal{Y}_{-n} := \mathcal{Y}_n^{\vee} = \mathrm{Hom}_{\mathbb{K}}(\mathcal{Y}_n, \mathbb{K}).$$

The negative index records categorical duality, rather than extending the preceding construction to negative n. Also set $\mathcal{Y}_0 := \mathbb{K}$, concentrated in even parity and bidegree $(0, 0)$, with the counit action. Thus $\mathcal{Y}_a$ is defined for every $a \in \mathbb{Z}$, with $\dim_{\mathbb{K}} \mathcal{Y}_a = 4|a|$ for $a \neq 0$ and $\dim_{\mathbb{K}} \mathcal{Y}_0 = 1$. For homogeneous $f \in \mathcal{Y}_{-n}$, h in the paired double, and $y \in \mathcal{Y}_n$, its right action is

$$\langle f \triangleleft h, y \rangle = (-1)^{|h|(|f|+1)} \langle f, y \triangleleft S^{-1}(h) \rangle. \tag{36}$$

This is the unique action for which evaluation is a right-module morphism. If $\{e_a\}$ is a homogeneous basis and $\{e^a\}$ its homogeneous dual basis, we use

$$\deg e^a = -\deg e_a, \qquad \mathrm{ev}_n(f \otimes y) = f(y), \qquad \mathrm{coev}_n(1) = \sum_a e_a \otimes e^a. \tag{37}$$

The sign in (36) and the use of S^{-1} are forced by the left-dual convention. With them, ev_n and coev_n are module morphisms and satisfy the two super snake identities.

Taking the left categorical dual of the module identification (32) gives, for $n > 0$,

$$\mathcal{Y}_{-n} \cong (W^{\mathrm{ao}}_{n,\beta})^{\vee} \boxtimes (L_{n,\beta})^{\vee}. \tag{38}$$

This is an identification of modules for the twisted paired double. By (26), its left dual agrees with the one defined in (36); no additional twist duality gauge is required. The root factor is the $4n$-dimensional simple dual module, while the central factor is one-dimensional with characters

$$\mathsf{c}_1 \mapsto 2n(\beta+1), \qquad \mathsf{c}_2 \mapsto -2n\beta.$$

Thus negative colors are obtained by duality, without substituting a negative integer into the highest-weight construction of Section 5.

For every integer color a, set

$$U_a = Q^{-2a\beta H_1 + 2a(\beta+1)H_2}. \tag{39}$$

Then $U_{a+b} = U_a U_b$, $U_{-a} = U_a^{-1}$, and $U_0 = 1$. For $n > 0$, the inverse charge on $\mathcal{Y}_{-n}$ agrees with the antipode in (36). If $x \in \mathcal{Y}_a$ has bidegree $p = (p_1, p_2)$, then

$$x \triangleleft H_1 = -p_2 x, \qquad x \triangleleft H_2 = -p_1 x, \qquad x \triangleleft U_b = Q^{-2b(\beta+1)p_1 + 2b\beta p_2} x.$$

These formulas hold for positive colors by (12), for negative colors by duality and the negative-degree convention, and for zero by the counit action.

Let

$$\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_r), \qquad \alpha_i \in \mathbb{Z}, \qquad \mathcal{Y}_{\boldsymbol{\alpha}} = \mathcal{Y}_{\alpha_1} \otimes \cdots \otimes \mathcal{Y}_{\alpha_r}.$$

Extend χ to nonzero bihomogeneous elements of $\bigoplus_{a\in\mathbb{Z}} \mathcal{Y}_a$ by the same formula (8), using the negative degrees on the dual from (37). This covers pairs from any two colors; parity is still the sum of the two degree coordinates modulo two. For bihomogeneous $x_i \in \mathcal{Y}_{\alpha_i}$, define the oriented staircase gauge

$$\begin{aligned} G^{\mathrm{or}}_{\boldsymbol{\alpha}}(x_1 \otimes \cdots \otimes x_r) = {} & \left(\prod_{1 \le i < j \le r} (-1)^{|x_i||x_j|} \chi(x_j, x_i) \right) \\ & \cdot \bigotimes_{i=1}^{r} \left(x_i \triangleleft \prod_{j=i+1}^{r} U_{\alpha_j} \right). \end{aligned} \tag{40}$$

All charges lie in the same commutative torus, so the products in (40) are unambiguous. Each gauge is diagonal with nonzero powers of Q as eigenvalues, hence invertible. A zero-colored tensor factor contributes no pairwise coefficient or charge, so the gauge is compatible with the tensor-unit identifications.

For a right–right super Yetter–Drinfeld module, the checked braiding is

$$c^{\mathrm{s}}_n(a \otimes b) = \sum (-1)^{|a||b_{(0)}|} b_{(0)} \otimes (a \triangleleft b_{(1)}). \tag{41}$$

For $y \in \mathcal{Y}_n$ of bidegree (a, b), define

$$\varphi_n(y) = Q^{-2n(\beta+1)a + 2n\beta b} y = y \triangleleft U_n.$$

The equality follows from (12) and (11). For the all-positive word $(n, \dots, n)$, the action on the i-th factor in (40) is $y_i \triangleleft U_n^{r-i} = \varphi_n^{r-i}(y_i)$, with $\varphi_n^0 = \mathrm{id}$.

For $\alpha, \gamma \in \mathbb{Z}$, let

$$c^{\mathrm{s}}_{\alpha,\gamma} : \mathcal{Y}_\alpha \otimes \mathcal{Y}_\gamma \longrightarrow \mathcal{Y}_\gamma \otimes \mathcal{Y}_\alpha$$

be the canonical braiding of the right-module category; in particular, $c^{\mathrm{s}}_{n,n} = c^{\mathrm{s}}_n$ for $n > 0$. Define the colored T-crossings by

$$T_{\alpha,\gamma} = G^{\mathrm{or}}_{(\gamma,\alpha)}\, c^{\mathrm{s}}_{\alpha,\gamma}\, (G^{\mathrm{or}}_{(\alpha,\gamma)})^{-1}. \tag{42}$$

For each $n > 0$, (42) includes $T_{n,n}$, $T_{-n,n}$, $T_{n,-n}$, and $T_{-n,-n}$. The identification $T_{n,n} = T_n$ with the original Garoufalidis–Kashaev operator is verified below.

Proposition 6.1 (Oriented braid-groupoid transport)**.** *For $r \geq 2$ and $1 \leq i < r$, let $s_i\boldsymbol{\alpha}$ be obtained by exchanging the i-th and $(i+1)$-st entries. Then*

$$(T_{\boldsymbol{\alpha}})_i = G^{\mathrm{or}}_{s_i\boldsymbol{\alpha}}\,(c^{\mathrm{s}}_{\boldsymbol{\alpha}})_i\,(G^{\mathrm{or}}_{\boldsymbol{\alpha}})^{-1}. \tag{43}$$

Under the identification of Appendix A, $T_{n,n} = T_n$ for every $n > 0$. Consequently, the operators in (42) extend the Garoufalidis–Kashaev braid operators and satisfy the colored oriented braid-groupoid relations. Crossings involving color zero are the canonical tensor-unit maps.

Proof. We first identify the positive crossing with the original operator.

Let $a, b \in \mathcal{Y}_n$ be bihomogeneous. For the $B_{\mathbb{K}}$-action, we use the same symbol $\triangleleft$ for the Garoufalidis–Kashaev twisted right-adjoint action $a \triangleleft h = \lambda_R(a \otimes h)$; see Appendix A.2. Decompose $\Delta_B(b) = \sum b^{(1)} \otimes b^{(2)}$ into bihomogeneous components. Expanding (41) using (13) gives

$$c^{\mathrm{s}}_n(a \otimes b) = \sum (-1)^{|b^{(2)}|(|b^{(1)}|+|a|)} \chi(b^{(2)}, b^{(1)})^{-1} \chi(b^{(1)}, a) \chi(b^{(2)}, a)\, b^{(1)} \otimes (\varphi_n(a) \triangleleft b^{(2)}).$$

To compute the conjugation, first apply

$$(G^{\mathrm{or}}_{(n,n)})^{-1}(a \otimes b) = (-1)^{|a||b|} \chi(b, a)^{-1} \varphi_n^{-1}(a) \otimes b.$$

This cancels φ_n in the right-adjoint term above. The resulting second tensor factor has degree $\deg a + \deg b^{(2)}$. Applying $G^{\mathrm{or}}_{(n,n)}$ therefore contributes the scalar

$$(-1)^{(|a|+|b^{(2)}|)|b^{(1)}|} \chi(a, b^{(1)}) \chi(b^{(2)}, b^{(1)})$$

and applies φ_n to the first tensor factor. Since $|b| = |b^{(1)}| + |b^{(2)}|$, all parity signs cancel. Also, $\chi(b, a) = \chi(b^{(1)}, a)\chi(b^{(2)}, a)$. Thus

$$\begin{aligned} &G^{\mathrm{or}}_{(n,n)} c^{\mathrm{s}}_n (G^{\mathrm{or}}_{(n,n)})^{-1}(a \otimes b) \\ &\quad = \sum \chi(a, b^{(1)})\, \varphi_n(b^{(1)}) \otimes (a \triangleleft b^{(2)}) \\ &\quad = T_n(a \otimes b), \end{aligned}$$

where the last equality is (74) under the identification of Appendix A.

For an arbitrary oriented word, the pairwise coefficients involving any other strand cancel because the adjacent braiding preserves the total bidegree of its two inputs. The charge factors cancel as well: a charge on a later strand acts on both adjacent inputs through its group-like coproduct and commutes through the module braiding, while an earlier strand sees the unchanged product $U_{\alpha_i} U_{\alpha_{i+1}}$. The remaining factors are exactly the two-strand gauges in (42). This proves (43), including the all-positive case. Conjugating the canonical braid relations by the boundary gauges proves the braid-groupoid relations. □

Set $G^{\mathrm{or}}_{\varnothing} = \mathrm{id}_{\mathbb{K}}$. For each $n > 0$, use the dual pair $(\mathcal{Y}_{-n}, \mathcal{Y}_n)$ with evaluation and coevaluation from (37). Its transported cups and caps are

$$\begin{aligned} \mathrm{ev}^T_n &= \mathrm{ev}_n (G^{\mathrm{or}}_{(-n,n)})^{-1}, \\ \mathrm{coev}^T_n &= G^{\mathrm{or}}_{(n,-n)}\, \mathrm{coev}_n\,. \end{aligned} \tag{44}$$

The comparison with standard duality is given in the next subsection.

6.2 Comparison with standard duality

The equality $T_{n,n} = T_n$ identifies the positive crossing. The mixed crossings, however, use the transported duality maps (44). We now compare them with the standard partial transposes used in [6, Section 2]. Throughout this comparison $n > 0$, and tensor products of the underlying linear maps are ordinary tensor products.

For y of bidegree (a, b), define

$$A_n(y) = Q^{2ab-2n(\beta+1)a+2n\beta b} y.$$

Its ordinary dual A_n^* is even and invertible. Direct substitution in (40) gives

$$\mathrm{ev}_n = \mathrm{ev}_n^T((A_n^*)^{-1} \otimes \mathrm{id}), \qquad \mathrm{coev}_n = (\mathrm{id} \otimes A_n^*)\,\mathrm{coev}_n^T\,. \tag{45}$$

Thus A_n^* identifies the transported negative object with the standard dual of $\mathcal{Y}_n$. The underlying vector space is the same; it is the evaluation and coevaluation that differ.

To specify the transpose convention, put $V = \mathcal{Y}_n$. For a linear map $f : V \otimes V \to V \otimes V$, define

$$\widetilde{f} = (\mathrm{ev}_n \otimes \mathrm{id}_V \otimes \mathrm{id}_{V^*})(\mathrm{id}_{V^*} \otimes f \otimes \mathrm{id}_{V^*})(\mathrm{id}_{V^*} \otimes \mathrm{id}_V \otimes \mathrm{coev}_n). \tag{46}$$

Its source is $V^* \otimes V$ and its target is $V \otimes V^*$. In an ordinary dual basis this reads

$$\widetilde{f}(e^a \otimes e_b) = \sum_{c,d} f_{bd}^{ac}\, e_c \otimes e^d, \qquad f(e_b \otimes e_d) = \sum_{a,c} f_{bd}^{ac} e_a \otimes e_c.$$

This is the partial transpose of [6, Section 2.1]; no super flip is inserted in this contraction. In particular, the operation is linear in f. Taking an inverse crossing and taking a partial transpose are distinct operations.

For the double-negative crossings, bend a second strand, following [7, Section 2, equations (10)–(11)]:

$$\widetilde{\widetilde{f}} = (\mathrm{ev}_n \otimes \mathrm{id}_{V^*} \otimes \mathrm{id}_{V^*})(\mathrm{id}_{V^*} \otimes \widetilde{f} \otimes \mathrm{id}_{V^*})(\mathrm{id}_{V^*} \otimes \mathrm{id}_{V^*} \otimes \mathrm{coev}_n) : V^* \otimes V^* \longrightarrow V^* \otimes V^*.$$

Here the second contraction uses the source and target of $\widetilde{f}$ specified above. Explicitly,

$$\widetilde{\widetilde{f}}(e^a \otimes e^c) = \sum_{b,d} f_{db}^{ca}\, e^b \otimes e^d.$$

Thus this double partial transpose is the full dual with the tensor factors reversed; no super flip is inserted.

Lemma 6.2 (Comparison of all oriented crossings)**.** *The positive-positive crossings are* $T_{n,n}^{\pm 1} = T_n^{\pm 1}$. *The remaining six crossings satisfy*

$$\widetilde{T_n^{-1}} = (\mathrm{id}_V \otimes A_n^*) T_{-n,n}((A_n^*)^{-1} \otimes \mathrm{id}_V), \tag{47}$$

$$(\widetilde{T_n})^{-1} = (A_n^* \otimes \mathrm{id}_V) T_{n,-n}(\mathrm{id}_V \otimes (A_n^*)^{-1}), \tag{48}$$

$$(\widetilde{T_n^{-1}})^{-1} = (A_n^* \otimes \mathrm{id}_V) T_{-n,n}^{-1}(\mathrm{id}_V \otimes (A_n^*)^{-1}), \tag{49}$$

$$\widetilde{T_n} = (\mathrm{id}_V \otimes A_n^*) T_{n,-n}^{-1}((A_n^*)^{-1} \otimes \mathrm{id}_V), \tag{50}$$

$$\widetilde{\widetilde{T_n}} = (A_n^* \otimes A_n^*) T_{-n,-n}((A_n^*)^{-1} \otimes (A_n^*)^{-1}), \tag{51}$$

$$\widetilde{\widetilde{T_n^{-1}}} = (A_n^* \otimes A_n^*) T_{-n,-n}^{-1}((A_n^*)^{-1} \otimes (A_n^*)^{-1}). \tag{52}$$

In the first and fourth identities the source and target are $V^*\otimes V$ and $V\otimes V^*$; in the second and third they are $V\otimes V^*$ and $V^*\otimes V$. The last two are endomorphisms of $V^*\otimes V^*$. Hence, after changing the negative-edge identification by A_n^*, all oriented crossings agree with the standard-duality operators prescribed in [6, Section 2] and [7, Section 2, equations (16)–(19)].

Proof. First, the transported mate identities may be read using ordinary local tensor products for the cup and cap contractions at issue here. Indeed, a cup inserts degrees $p, -p$ and charges U_n, U_{-n}. In the pairwise gauge, its coefficients with every external strand cancel, since χ is multiplicative in the degrees and $U_nU_{-n}=1$. Only the internal two-strand factor remains, which is exactly coev_n^T; the same argument applies to a cap. Consequently, tensoring a local cup or cap with identity maps on external factors is compatible with conjugation by the full boundary gauges. Together with (43), this shows that each local crossing, cup, or cap is the corresponding canonical map conjugated by its input and output boundary gauges. These gauges cancel on composition, so the canonical snake and braiding evaluation identities give the transported identities using ordinary tensor products.

The braiding evaluation identity expresses $T_{-n,n}$ as the contraction (46) of T_n^{-1}, with $\mathrm{ev}_n, \mathrm{coev}_n$ replaced by $\mathrm{ev}_n^T, \mathrm{coev}_n^T$. Substituting (45) into that contraction gives

$$T_{-n,n} = (\mathrm{id}_V \otimes (A_n^*)^{-1})\widetilde{T_n^{-1}}(A_n^* \otimes \mathrm{id}_V),$$

which proves (47). The other evaluation identity expresses $(T_{n,-n})^{-1}$ as the same transported contraction of T_n. Substitution gives

$$(T_{n,-n})^{-1} = (\mathrm{id}_V \otimes (A_n^*)^{-1})\widetilde{T_n}(A_n^* \otimes \mathrm{id}_V).$$

Inverting proves (48) and also establishes the required invertibility of both partial transposes. Taking inverses gives (49) and (50).

Bending both strands gives the categorical dual of the positive crossing. The dual of a tensor product reverses its factor order, so, in the ordinary dual bases fixed above, this map is exactly $\widetilde{\widetilde{T_n}}$, as the displayed coefficient formula shows. Each negative boundary factor is identified with the standard dual by A_n^*, giving (51). Taking its inverse gives (52), since the reversed tensor dual commutes with inversion:

$$\widetilde{\widetilde{T_n^{-1}}} = (\widetilde{\widetilde{T_n}})^{-1}.$$

□

For clarity, temporarily write $J_{T_n}^{\mathrm{tr}}(D)$ for the normal long-knot operator computed with the transported crossings and cups, and $J_{T_n}^{\mathrm{GK}}(D)$ for the standard-duality state sum of [6, Section 2.2]. Change the identification on every negative edge by A_n^*, leaving positive edges unchanged. The lemma and (45) convert all local oriented crossings, cups, and caps to their standard-duality counterparts. On every internal edge the identification and its inverse cancel. Both external edges are positive, so

$$J_{T_n}^{\mathrm{tr}}(D) = J_{T_n}^{\mathrm{GK}}(D). \tag{53}$$

We henceforth denote this common operator by $J_{T_n}(D)$. This is equality of the assembled operators, not equality of the individual mixed-crossing matrices before changing the dual identification.

For later use, the same bending argument applies to the root braiding $R'_{n,\beta}$: its standard partial transposes and those of its inverse are invertible. For any nonzero scalar μ, linearity and inversion give

$$\widetilde{\mu R} = \mu \widetilde{R}, \qquad \widetilde{(\mu R)^{-1}} = \mu^{-1}\widetilde{R^{-1}}, \qquad (\widetilde{\mu R})^{-1} = \mu^{-1}(\widetilde{R})^{-1}.$$

The inverse scalar in a mixed crossing therefore comes from taking an inverse, not from partial transpose itself.

7 Long-knot transport and the Links–Gould comparison

7.1 Compatibility with the root–center twist

The preceding gauge connects T_n with the untwisted braiding. We now include the tensorator of $\mathcal{F}_{\mathrm{rc}}$ on the same oriented words. If $x \in \mathcal{Y}_\alpha$ has bidegree $p = (p_1, p_2)$, then

$$\begin{aligned} x \triangleleft \mathsf{h}_1 &= (\alpha(\beta+1) - p_2)x, \quad & x \triangleleft \mathsf{h}_2 &= (-\alpha\beta - p_1)x, \\ x \triangleleft \mathsf{c}_1 &= -2\alpha(\beta+1)x, & x \triangleleft \mathsf{c}_2 &= 2\alpha\beta x. \end{aligned} \tag{54}$$

For $\alpha, \gamma \in \mathbb{Z}$, define

$$\begin{aligned} \Gamma_{\alpha|\gamma}(p,q) = \frac{1}{2}\big[&\gamma((\beta+1)p_1 - \beta p_2) \\ &+ \alpha(\beta q_2 - (\beta+1)q_1)\big]. \end{aligned} \tag{55}$$

Let $\mathcal{F}_{\mathrm{rc}}^{[r]}$ be the coherent r-fold tensorator of the normalized twist and put

$$\mathsf{F}^{\mathrm{or}}_{\boldsymbol{\alpha}} = \rho_{\mathcal{Y}_{\boldsymbol{\alpha}}}(\mathcal{F}_{\mathrm{rc}}^{[r]}). \tag{56}$$

Equations (24) and (54) give the explicit diagonal action

$$\mathsf{F}^{\mathrm{or}}_{\boldsymbol{\alpha}}(x_1 \otimes \cdots \otimes x_r) = Q^{\sum_{i<j} \Gamma_{\alpha_i|\alpha_j}(p_i,p_j)} x_1 \otimes \cdots \otimes x_r. \tag{57}$$

The negative-color signs in this formula come from categorical duality, not from an additional convention.

Let $(c^{\mathrm{rc}}_{\boldsymbol{\alpha}})_i$ be the braiding of the twisted paired double. Monoidal transport under a Drinfeld twist gives

$$(c^{\mathrm{rc}}_{\boldsymbol{\alpha}})_i = (\mathsf{F}^{\mathrm{or}}_{s_i\boldsymbol{\alpha}})^{-1}(c^{\mathrm{s}}_{\boldsymbol{\alpha}})_i \mathsf{F}^{\mathrm{or}}_{\boldsymbol{\alpha}}. \tag{58}$$

Define the combined gauge

$$\mathsf{G}^{\mathrm{rc,or}}_{\boldsymbol{\alpha}} = G^{\mathrm{or}}_{\boldsymbol{\alpha}} \mathsf{F}^{\mathrm{or}}_{\boldsymbol{\alpha}}. \tag{59}$$

Substitution in (43) yields

$$(T_{\boldsymbol{\alpha}})_i = \mathsf{G}^{\mathrm{rc,or}}_{s_i\boldsymbol{\alpha}} (c^{\mathrm{rc}}_{\boldsymbol{\alpha}})_i (\mathsf{G}^{\mathrm{rc,or}}_{\boldsymbol{\alpha}})^{-1}. \tag{60}$$

This formula relates the integer-colored T-crossings to the braiding of the twisted paired double through the combined gauge.

For each $n > 0$, the twist leaves the standard evaluation and coevaluation unchanged:

$$\mathrm{ev}_n \, \mathsf{F}^{\mathrm{or}}_{(-n,n)} = \mathrm{ev}_n, \qquad (\mathsf{F}^{\mathrm{or}}_{(n,-n)})^{-1} \, \mathrm{coev}_n = \mathrm{coev}_n \, . \tag{61}$$

Indeed, their paired degrees give $\Gamma_{-n|n}(-p,p) = \Gamma_{n|-n}(p,-p) = 0$, in agreement with (26). Using (59) and (44), the combined gauge therefore satisfies

$$\mathrm{ev}_n (\mathsf{G}^{\mathrm{rc,or}}_{(-n,n)})^{-1} = \mathrm{ev}_n^T, \tag{62}$$

$$\mathsf{G}^{\mathrm{rc,or}}_{(n,-n)} \, \mathrm{coev}_n = \mathrm{coev}_n^T \, . \tag{63}$$

These identities provide the local crossing, cup, and cap comparisons needed for long-knot transport.

7.2 The normal long-knot operator

Let D be a normal long-knot diagram and let $J_R(D)$ be its endomorphism-valued state sum using standard duality as in [6, Section 2]. For $R = T_n$, its agreement with the transported construction is (53). Composing the local comparison at every crossing, cup, and cap gives the following global statement.

Theorem 7.1 (Long-knot transport)**.** *For every normal long-knot diagram D,*

$$\Phi_n J_{T_n}(D)\Phi_n^{-1} = (\lambda_{n,\beta}^{\mathrm{cen}})^{\mathrm{wr}(D)} J_{R'_{n,\beta}}(D). \tag{64}$$

In particular, the two operators are conjugate whenever D is balanced, that is, $\mathrm{wr}(D) = 0$.

Proof. By Lemma 6.2, (53), and the combined local transport identities above, the boundary gauges cancel when the elementary pieces of D are composed. The remaining external identification is Φ_n. Root–center factorization contributes $\lambda_{n,\beta}^{\mathrm{cen}}$ at each positive crossing and its inverse at each negative crossing, by the scalar rules in Section 6.2. Their product is $(\lambda_{n,\beta}^{\mathrm{cen}})^{\mathrm{wr}(D)}$, proving the formula. □

7.3 Balanced representatives

For the knot comparison below, choose a balanced normal diagram D representing $\mathcal{K}$. Such a representative is obtained by composing a normal diagram with Kashaev's correction $\xi^{-\mathrm{wr}(D)/2}$: normal diagrams have even writhe and $\mathrm{wr}(\xi^k) = 2k$ [7, Section 1, Remark 3, and Section 2, Theorem 1]. The state sum on this corrected diagram is the normalized knot operator. In the remainder of this section, D denotes a balanced normal representative, so the central scalar in Theorem 7.1 is one.

7.4 Scalarity

The operator $J_{R'_{n,\beta}}(D)$ is an even $U_\hbar^{\Pi_{\mathrm{ao}}}(\mathfrak{sl}(2|1))$-module endomorphism of $W_{n,\beta}^{\mathrm{ao}}$: braidings, their inverses, cups, caps, tensor products, and compositions are all module morphisms.

Proposition 7.2.

$$\mathrm{End}_{U_\hbar^{\Pi_{\mathrm{ao}}}(\mathfrak{sl}(2|1))}(W_{n,\beta}^{\mathrm{ao}}) = \mathbb{K}\,\mathrm{id}\,.$$

Consequently, there is a unique scalar $LG'_{n,\mathcal{K}}(Q^{n\beta}, Q)$ such that

$$J_{R'_{n,\beta}}(D) = LG'_{n,\mathcal{K}}(Q^{n\beta}, Q)\,\mathrm{id}\,.$$

Proof. An endomorphism sends the one-dimensional highest-weight space to itself. It therefore acts on $w_{n,\beta}$ by a scalar. Since $W_{n,\beta}^{\mathrm{ao}}$ is generated by $w_{n,\beta}$, the same scalar acts on the whole module. □

Combining Theorem 7.1 and Proposition 7.2, and recalling that $V_{n,\mathcal{K}}$ is the $(1,1)$-entry of $J_{T_n}(D)$, gives

$$V_{n,\mathcal{K}}(Q^{2n\beta+n}, Q^2) = LG'_{n,\mathcal{K}}(Q^{n\beta}, Q). \tag{65}$$

In fact, the argument proves the stronger scalar identity

$$J_{T_n}(D) = LG'_{n,\mathcal{K}}(Q^{n\beta}, Q)\,\mathrm{id}_{\mathcal{Y}_n}$$

for balanced normal diagrams. Thus scalarity is a theorem rather than an input.

7.5 Comparison with standard Links–Gould conventions

It remains to compare the root-module scalar with the standard Links–Gould normalization.

Theorem 7.3. *For every oriented knot $\mathcal{K}$,*

$$LG'_{n,\mathcal{K}}(Q^{n\beta}, Q) = LG^{(n)}_{\mathcal{K}}(Q^{-n\beta}, Q^{-1}) = LG^{(n)}_{\overline{\mathcal{K}}}(Q^{n\beta}, Q). \tag{66}$$

Proof. The odd reflection (34) identifies the coloring object with $\Pi V_0(n-1, n\beta)$. Independence of the Drinfeld–Jimbo quantization from the simple-root system, together with the quantum-group/Kontsevich comparison, identifies the corresponding normalized rigid-tangle scalars [9, 11]. The parity shift contributes only a writhe factor, which is one on the balanced representatives used here.

Standard Links–Gould conventions use left modules with braiding $\tau^{\mathrm{s}}(\pi\otimes\pi)(\mathcal{R})$, whereas (5) uses $\mathcal{R}^{-1}$. The right-to-left conversion therefore reverses the crossings, producing the simultaneous substitutions $(Q^{n\beta}, Q) \mapsto (Q^{-n\beta}, Q^{-1})$, or, equivalently, taking the mirror with parameters fixed. This convention is uniform in n and agrees with [8, Theorem 1.5], proving (66). □

Proof of Theorem 1.1. Substitute (66) into (65). The scalar-identity assertion follows from Proposition 7.2. □

7.6 Concluding remarks

The staircase gauge identifies the Nichols braid operator with the super Yetter–Drinfeld braiding, while the root–center twist separates the quantum-group contribution from the scalar $Q^{2n^2\beta(\beta+1)}$. Balanced normalization removes this scalar. The remaining long-knot operator acts on a simple typical module, so its $(1,1)$-entry determines the entire operator.

Appendices

A Comparison with the Garoufalidis–Kashaev conventions

We record the parameter and action conventions relating our construction to [6, Theorem 3.6 and Definition 7.5].

A.1 Parameters and the space Y_n

To distinguish the notation of [6] from ours, add a superscript GK to its parameters. The required specialization is

$$q^{\mathrm{GK}} = Q^2, \qquad q_{12}^{\mathrm{GK}} = q_{21}^{\mathrm{GK}} = -Q, \qquad t^{\mathrm{GK}} = Q^{2n\beta+n} = t. \tag{67}$$

Indeed, the diagonal braiding determined by $q_{11}^{\mathrm{GK}} = q_{22}^{\mathrm{GK}} = -1$ and $q_{12}^{\mathrm{GK}} = q_{21}^{\mathrm{GK}} = -Q$ is exactly the braiding τ_χ determined by (8). The parametrization following [6, Proposition 7.4] therefore gives

$$\begin{aligned} t_1^{\mathrm{GK}} &= \frac{1}{(q^{\mathrm{GK}})^{n/2} t^{\mathrm{GK}}} = Q^{-2n(\beta+1)}, \\ t_2^{\mathrm{GK}} &= \frac{t^{\mathrm{GK}}}{(q^{\mathrm{GK}})^{n/2}} = Q^{2n\beta}. \end{aligned} \tag{68}$$

In particular, $t_1^{\mathrm{GK}} t_2^{\mathrm{GK}} (q^{\mathrm{GK}})^n = 1$. The coefficient of the invariant vector in [6, (110)–(113)] is

$$\alpha^{\mathrm{GK}} = t_2^{\mathrm{GK}} (-q_{12}^{\mathrm{GK}})^n = Q^{2n\beta+n} = t. \tag{69}$$

Consequently,

$$v_{n,\alpha^{\mathrm{GK}}} = (x_1 x_2)^n + \alpha^{\mathrm{GK}} (x_2 x_1)^n = v_n,$$

and the basis in [6, Proposition 7.4] becomes exactly the basis (10). Thus, after the specialization (67) and extension of scalars to $\mathbb{K}$, there is a canonical identification

$$Y_n^{\mathrm{GK}} \otimes_{\mathbb{F}} \mathbb{K} \cong \mathcal{Y}_n \tag{70}$$

inside $B_{\mathbb{K}}$. Since $Q = e^{\hbar}$, the parameter $q^{\mathrm{GK}} = Q^2$ is not a root of unity in the formal setting. Moreover, $\beta \neq -1, 0$ gives respectively $t_1^{\mathrm{GK}} \neq 1$ and $t_2^{\mathrm{GK}} \neq 1$, so the hypotheses of [6, Proposition 7.4] hold.

A.2 The right action and the original definition of T_n

The map φ_n used in Section 6.1 is the restriction of the scaling automorphism of the braided Hopf algebra $B_{\mathbb{K}}$ determined by

$$\varphi_n(x_1) = Q^{-2n(\beta+1)} x_1, \qquad \varphi_n(x_2) = Q^{2n\beta} x_2. \tag{71}$$

We also write

$$a \triangleleft h = \lambda_R(a \otimes h), \qquad a, h \in B_{\mathbb{K}},$$

for the twisted right-adjoint action of [6, Theorem 3.5]. Its restriction to $\mathcal{Y}_n$ agrees with the action above when the acting element belongs to $B_{\mathbb{K}}$; Cartan and paired-double actions retain their previously specified domains. On the generators it is

$$\begin{aligned} a \triangleleft x_1 &= a x_1 - Q^{-2n(\beta+1)} \chi(a, x_1) x_1 a, \\ a \triangleleft x_2 &= a x_2 - Q^{2n\beta} \chi(a, x_2) x_2 a, \end{aligned} \tag{72}$$

which is precisely the restriction used in (12).

Under (70), the operator called T_n in [6, Theorem 3.6 and Definition 7.5] is

$$T_n^{\mathrm{GK}} = (\varphi_n \otimes \lambda_R)(\tau_\chi \otimes \mathrm{id})(\mathrm{id} \otimes \Delta_B)\big|_{\mathcal{Y}_n \otimes \mathcal{Y}_n}. \tag{73}$$

Thus, if a, b are bihomogeneous and

$$\Delta_B(b) = \sum b^{(1)} \otimes b^{(2)} \quad \text{in bihomogeneous components},$$

then

$$T_n^{\mathrm{GK}}(a \otimes b) = \sum \chi(a, b^{(1)})\, \varphi_n(b^{(1)}) \otimes (a \triangleleft b^{(2)}). \tag{74}$$

This fixes both the tensor order and the right-action convention.

A.3 Parameter dictionary

Symbol	Meaning	Relation
Q	formal quantum parameter	$Q = e^{\hbar}$
β	highest-weight parameter	$\beta \neq 0, -1$
α	standard Links–Gould weight	$\alpha = n\beta$
t	first V_n-parameter	$t = Q^{2n\beta+n}$
q	second V_n-parameter	$q = Q^2$

B First root degrees of the canonical tensor

For reference, the inverses of the pairing matrices are

$$M_{2r+1}^{-1} = \frac{(-1)^r}{P_r(Q)} I_2,$$
$$M_{2r}^{-1} = \frac{(-1)^r}{P_r(Q)} \begin{pmatrix} 1 & Q^{-r} \\ Q^{-r} & 1 \end{pmatrix}.$$

The super right-dual convention multiplies degree m by $(-1)^m$. Thus the signed root coevaluation begins with

$$1 - x_1 \otimes F_1 - x_2 \otimes F_2 + \text{terms of root degree at least two.}$$

Its inverse, the standard root factor of the universal R-matrix, begins with

$$1 + x_1 \otimes F_1 + x_2 \otimes F_2 + \text{terms of root degree at least two.}$$

These signs are forced simultaneously by the right-dual evaluation and the cross relation $x_i F_i + F_i x_i = Q^{H_i} - Q^{\widehat{H}_i}$.

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